Stochastic Processes

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Abstract

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1 Introduction to Stochastic Processes

1.1 Basic Terminology

Definition 1.1.1 (Stochastic Processes) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let (S, Σ) be a measurable space. A stochastic process is a collection of random variables

$$\{X(t) : \Omega \to S \mid t \in T\}$$

indexed by some set T. In this case we write the stochastic process as $X : \Omega \times T \to S$.

Definition 1.1.2 (Terminology) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let (S, Σ) be a measurable space. Let $X : \Omega \times T \to S$ be a stochastic process.

- Define the state space of the stochastic process to be S.
- Write $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$.

Definition 1.1.3 (Discrete Time Stochastic Processes) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let (S, Σ) be a measurable space. Let $X : \Omega \times T \to S$ be a stochastic process. We say that the stochastic process has discrete time if $T \subseteq \mathbb{N}$.

Definition 1.1.4 (Filtrations)
Let T be an indexing set. Let $\{\mathcal{F}_t \mid t \in T\}$ be a family of σ-algebras. We say that $\mathcal{F}$ is a filtration if $\mathcal{F}_t \leq \mathcal{F}_s$ for $t \leq s$.

Definition 1.1.5 (Right Continuous Filtrations)
Let T Be an indexing set. Let $\{\mathcal{F}_t \mid t \in T\}$ be a family of σ-algebras that is a filtration. We say that the filtration is right continuous if

$$\mathcal{F}_t = \bigcap_{s>t} \mathcal{F}_s$$

Definition 1.1.6 (Stochastic Bases (Filtered Probability Spaces))
Let $I \subseteq \mathbb{R}$. A stochastic basis is a probability space $(\Omega, \mathcal{F}, P)$ together with a filtration $\{\mathcal{F}_t \mid t \in I\}$.

Definition 1.1.7 (Adapted Stochastic Process)
Let $T \subseteq \mathbb{R}$. Let $(\Omega, \mathcal{F}, P)$ be a stochastic basis with filtration $\{\mathcal{F}_t \mid t \in T\}$. Let $X : \Omega \times T \to \mathbb{R}$ be a stochastic process. We say that X is adapted to the filtration if X_t is $\mathcal{F}_t$ measurable for all $t \in T$.

The filtration here means that we learn more information as time passes. Indeed we assume that $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s \leq t$ so that as t increases, the random variable X_t has a bigger and bigger domain.

Right continuity means that information arrives continuously with no sudden jumps.

1.2 Hitting Time and Stopping Time

Definition 1.2.1 (Hitting Time)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let (S, Σ) be a measurable space. Let $X : \Omega \times T \to S$ be a stochastic process. Let $A \in \Sigma$ be a measurable set. Define the hitting time of X to be the random variable $H_A : \Omega \to T$ given by

$$H_A(\omega) = \inf\{t \in T \mid X(\omega, t) \in A\}$$

The hitting time measures the first time that a stochastic process enters A for the first time.

Recall that for a σ-algebra $\mathcal{F}$ and a subset $Y \subseteq \mathcal{F}$, $\sigma(Y)$ is the smallest σ-algebra that contains Y.

Definition 1.2.2 (Stopping Time)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let (S, Σ) be a measurable space. Let $X : \Omega \times T \to S$ be a stochastic process. Let $S : \Omega \to T$ be a random variable. We say that S is a stopping time if S has the property that
$$\{\omega \in \Omega \mid S(\omega) \leq t\} \subseteq \sigma(X_t)$$
for all $t \in T$.

In other words, events determined by S up to some time $t \in T$ is dictated entirely using the variables $X_0, \dots, X_t$.

Lemma 1.2.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let (S, Σ) be a measurable space. Let $X : \Omega \times T \to S$ be a stochastic process. Let $S : \Omega \to T$ be a random variable. Then S is a stopping time if and only if S has the property that
$$\{\omega \in \Omega \mid S(\omega) = t\} \subseteq \sigma(X_t)$$
for all $t \in T$.

Corollary 1.2.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let (S, Σ) be a measurable space. Let $X : \Omega \times T \to S$ be a stochastic process. Let $A \in \Sigma$ be a measurable set. Then H_A is a stopping time.

Theorem 1.2.5 (Début Theorem)

2 Discrete Time Markov Chains

2.1 Basic Definitions

Definition 2.1.1 (Discrete Time Markov Chains)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time stochastic process. We say that X is a Markov chain if

$$P(X_{n+1} = i_{n+1} \mid X_j = i_j \text{ for } 0 \leq j \leq n) = P(X_{n+1} = i_{n+1} \mid X_n = i_n)$$

for all $n \in \mathbb{N}$.

Definition 2.1.2 (State Space)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Define the state space of X to be $\text{im}(X)$.

Definition 2.1.3 (Homogeneous Markov Chains)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. We say that X is homogeneous if

$$P(X_{n+1} = j \mid X_n = i) = P(X_1 = j \mid X_0 = i)$$

Theorem 2.1.4 (Strong Markov Property)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Let $T : \Omega \to \mathbb{N}$ be a stopping time. Then $(X_{T+n})_{n \geq 0}$ is a discrete time Markov chain.

2.2 Communicating Classes

Definition 2.2.1 (Talking and Communicating States)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain.

- We say that $i \in I$ talks to $j \in I$ if $P(X_n = j \mid X_0 = i) > 0$ for some $n \in \mathbb{N}$.
- We say that $i, j \in I$ communicates if i talks to j and j talks to i.

Definition 2.2.2 (Communicating Class)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. A communicating class of X is an element of $\frac{\mathbb{R}}{\sim}$ where $x \sim y$ if x and y communicates.

Definition 2.2.3 (Closed Classes)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Let C be a communicating class. We say that C is closed if no states in the class talks to any state outside of the class.

Definition 2.2.4 (Irreducible Discrete Time Markov Chains)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. We say that X is irreducible if it has one communicating class.

2.3 Recurrent and Transient States

Definition 2.3.1 (Visit Counting Function)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Define the visit counting function of $i \in \mathbb{R}$ to be the random variable $V_i : \Omega \to \mathbb{R}$ given by

$$V_i = \sum_{j=0}^{\infty} 1_{X_j = i}$$

Definition 2.3.2 (Recurrent and Transient States)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain.

- We say that that $i \in I$ is a recurrent state if
$$P(V_i = \infty \mid X_0 = i) = 1$$
- We say that $i \in I$ is a transient state if
$$P(V_i = \infty \mid X_0 = i) = 0$$

Definition 2.3.3 (Passage Time)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Define the rth passage time of state $i \in \mathbb{R}$ to be the random variable $T_i^{(r)} : \Omega \to \mathbb{N}$ given by
$$T_i^{(r)}(\omega) = \inf\{n > T_i^{(r-1)} \mid X_n(\omega) = i\}$$
if $r \geq 1$ and $T_i^{(0)}(\omega) = 0$.

Definition 2.3.4 (Excursion Time)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Define the rth excusion time of state $i \in \mathbb{R}$ to be the random variable $S_i^{(r)} : \Omega \to \mathbb{N}$ given by
$$S_i^{(r)}(\omega) = \begin{cases} T_i^{(r)}(\omega) - T_i^{(r-1)}(\omega) & \text{if } T_i^{(r-1)}(\omega) < \infty \\ \infty & \text{otherwise} \end{cases}$$
for $r \geq 1$.

Lemma 2.3.5
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Let $r \geq 2$. Then the following are true.

- $S_i^{(r)}$ and X_m for $m \leq T_i^{(r-1)}$ are independent random variables.
- $P(S_i^{(r)} = n \mid ; T_i^{(r-1)} < \infty) = P(T_i = n \mid X_0 = i)$.

Proof
Since T_i is a stopping time, the first item follows from the strong Markov property.

By definition, we have
$$S_i^{(r)} = \inf\{n \geq 1 \mid X_{T_i^{(r-1)}+n}\}$$
so that $S_i^{(r)}$ is the first passage time of the Markov chain $(X_{T_i^{(r-1)}+n})$ to state i. ■

Lemma 2.3.6
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Then we have
$$P(V_i > r \mid X_0 = i) = P(T_i < \infty \mid X_0 = i)^r$$

Proof
Notice that
$$\{\omega \in \Omega \mid V_i(\omega) > r \text{ and } X_0(\omega) = i\} = \{\omega \in \Omega \mid T_i^{(r)} < \infty \text{ and } X_0(\omega) = i\}$$
We proceed by induction. When $r = 0$, we have $P(V_i > 0 \mid X_0 = i) = 1 = P(T_i < \infty \mid X_0 = i)^0$

trivially. Now suppose it is true for some $k \in \mathbb{N}$. Then we have

$$\begin{aligned}P(V_i > k+1 \mid X_0 = i) &= P(T_i^{(r+1)} < \infty \mid X_0 = i) \\ &= P(T_i^{(r)} < \infty \cap S_i^{r+1} < \infty \mid X_0 = i) \\ &= P(T_i^{(r)} < \infty \mid X_0 = i) P(T_i < \infty \mid X_0 = i) \\ &= P(T_i < \infty \mid X_0 = i)^r P(T_i < \infty \mid X_0 = i) \\ &= P(T_i < \infty \mid X_0 = i)^{r+1}\end{aligned}$$

∎

Proposition 2.3.7
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Let $i \in \mathbb{R}$ be a state. Then the following are true.

- If $P(T_i < \infty \mid X_0 = i) = 1$, then i is a recurrent state.
- If $P(T_i < \infty \mid X_0 = i) < 1$, then i is a transient state.

Proof
If $P(T_i < \infty \mid X_0 = i) = 1$, then we have

$$P(V_i = \infty \mid X_0 = i) = \lim_{n \to \infty} P(V_i > n \mid X_0 = i) = \lim_{n \to \infty} P(T_i < \infty \mid X_0 = i)^n = 1$$

Hence i is recurrent.

Now suppose that $P(T_i < \infty \mid X_0 = i) < 1$. Then we have

$$P(V_i = \infty \mid X_0 = i) = \lim_{n \to \infty} P(V_i > n \mid X_0 = i) = \lim_{n \to \infty} P(T_i < \infty \mid X_0 = i)^n = 0$$

Hence i is transient. ∎

Proposition 2.3.8
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Let $i, j \in \mathbb{R}$ be communicating states. Then i is recurrent if and only if j is recurrent.

Proposition 2.3.9
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Let C be a communicating class. Then the following are true.

- If C is recurrent, then C is closed.
- If C is finite and closed, then C is recurrent.

Definition 2.3.10 (Positive Recurrent)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \times \mathbb{N} \to \mathbb{R}$ be a discrete time Markov chain. Let $i \in \mathbb{R}$. We say that i is positive recurrent if $E[T_i \mid X_0 = i] < \infty$.

The definition means that we can always expect to visit state i again in finite time.

2.4 Discrete Time Markov Chains with Finite State Space

Definition 2.4.1 (Transition Matrix)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Define the transition matrix of X to be

$$T = \begin{pmatrix} P(X_1 = 1 \mid X_0 = 1) & \cdots & P(X_1 = n \mid X_0 = 1) \\ \vdots & \ddots & \vdots \\ P(X_1 = 1 \mid X_0 = n) & \cdots & P(X_1 = n \mid X_0 = n) \end{pmatrix}$$

Proposition 2.4.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Let T be the transition matrix of X. Then we have

$$P(X_n = j \mid X_0 = i) = (T^n)_{i,j}$$

where $(T^n)_{i,j}$ is the (i, j)th entry of T^n.

Proposition 2.4.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Then we have

$$P(X_n = j) = \sum_{i \in I} P(X_0 = i) P(X_1 = j \mid X_0 = i)^n$$

Proposition 2.4.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Let $A \subseteq I$. Let $(v_1, \dots, v_n)$ be the minimal non-negative solution of the system of linear equations:

$$x_i = \begin{cases} \sum_{j \in I} P(X_1 = j \mid X_0 = i) x_j & \text{if } i \notin A \\ 1 & \text{otherwise} \end{cases}$$

Then we have $v_i = P(H_A < \infty \mid X_0 = i)$.

Example 2.4.5
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain whose transition matrix is given by

$$\begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Let $A = \{2\}$. Then the system of linear equations of the hitting time is given by

$$x_1 = \frac{1}{2} x_2 + \frac{1}{2} x_3 \quad \text{and} \quad x_2 = 1 \quad \text{and} \quad x_3 = \frac{1}{3} x_1 + \frac{1}{3} x_2 + \frac{1}{3} x_4$$

The minimal solution is given by $(4/5, 1, 3/5, 0)$. Hence we have

$$P(H_A < \infty \mid X_0 = 1) = \frac{4}{5} \quad \text{and} \quad P(H_A < \infty \mid X_0 = 3) = \frac{3}{5}$$

Example 2.4.6
A pair of fair dice is rolled and its sum of the values are recorded. What is the probability that a sum of 12 appears before two consecutive sums of 7?

Answer We consider a Markov chain whose states are: the starting state, a sum of 7 appearing, two consecutive sums of 7 appearing and a sum of 12 appearing. Labelling them as $1, \dots, 4$ respectively, the transition matrix is given by

$$\begin{pmatrix} \frac{29}{36} & \frac{1}{6} & 0 & \frac{1}{36} \\ \frac{29}{36} & 0 & \frac{1}{6} & \frac{1}{36} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Let x_i be the probability that state 4 is reached before state 3 starting at state i. Similar to the linear equalities in hitting times, we have the system of linear equations

$$x_1 = \frac{29}{36}x_1 + \frac{1}{6}x_2 + \frac{1}{36}x_4 \quad \text{and} \quad x_2 = \frac{29}{36}x_1 + \frac{1}{6}x_3 + \frac{1}{36}x_4 \quad \text{and} \quad x_3 = 0 \quad \text{and} \quad x_4 = 1$$

Solving it gives $x_1 = \frac{7}{13}$. ⊛

Proposition 2.4.7
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Let $A \subseteq I$. Let $(v_1, \dots, v_n)$ be the minimal non-negative solution of the system of linear equations:

$$x_i = \begin{cases} 1 + \sum_{j \in I} P(X_1 = j \mid X_0 = i)x_j & \text{if } i \notin A \\ 0 & \text{otherwise} \end{cases}$$

Then we have $v_i = E[H_A \mid X_0 = i]$.

Definition 2.4.8 (Component Independent Simple Random Walk on $\mathbb{Z}^d$) A component independent simple random walk on $\mathbb{Z}^d$ for $d \in \mathbb{N}$ is defined as

$$X_{n+1} = X_n + Z_{n+1}$$

where $Z_{n+1} = (Z^1_{n+1}, \dots, Z^d_{n+1})$ with

$$Z^j_m = \begin{cases} 1 & \text{with probability } p_j \\ -1 & \text{with probabilty } q_j \end{cases}$$

where Z^j_m are independent for all j, m.

Proposition 2.4.9 Let $(X_n)_{n \geq 0}$ be a CISRW on $\mathbb{Z}^d$. If there exists $j \in \{1, \dots, d\}$ such that $p_j \neq \frac{1}{2}$ then (X_n) is transient.

Proposition 2.4.10 Let $(X_n)_{n \geq 0}$ be a CISRW on $\mathbb{Z}^d$ and $p_j = q_j = \frac{1}{2}$ for all j. Then
- If $d \leq 2$ then all states are recurrent
- If $d \geq 3$ then all states are transient

2.5 Stationary Distributions

Definition 2.5.1 (Stationary Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Let $v = (v_1, \dots, v_n) \in \mathbb{R}^n_{\geq 0}$ be a vector. Let T be the transition matrix of X. We say that v is a stationary distribution of X if the following are true.
- $\sum_{i=1}^n v_i = 1$.
- $Tv = v$.

Lemma 2.5.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Let v be a stationary distribution of X. Then the probability distribution function of X_i is given by
$$f_{X_n}(i) = v_i$$
for all $n \in \mathbb{N}$.

Proposition 2.5.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Let T be the transition matrix of X. Suppose that there exists $i \in I$ such that $((T^n)_{i,j})_{n\in\mathbb{N}}$ converges to some $v_j \in \mathbb{R}$ for all $j \in I$. Then $(v_1, \dots, v_n)$ is a stationary distribution of X.

Proposition 2.5.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $I = \{1, \dots, n\}$. Let $X : \Omega \times \mathbb{N} \to I$ be a discrete time Markov chain. Suppose that X is irreducible. Then the following are true.

- Every state is positive recurrent.
- $\left(\frac{1}{E[T_i \mid X_0=1]}, \dots, \frac{1}{E[T_i \mid X_0=n]}\right)$ is the unique stationary distribution of X.

3 Brownian Motion

Definition 3.0.1 (Brownian Motion) Let (Ω, E, P) be a probability space. Let (S, Σ) be a measurable space. Let $B : \Omega \times [0, T] \to S$ be continuous stochastic processes. We say that B is a Brownian motion if the following are true.

- $B_0 = 0$.
- For any $0 \leq t_1 < t_2 < t_3 \leq T$,
  $$B_{t_2} - B_{t_1} \quad \text{and} \quad B_{t_3} - B_{t_2}$$
  are independent variables.
- For any $0 \leq t_1 < t_2 \leq T$,
  $$B_{t_2} - B_{t_1} \sim N(0, t_2 - t_1)$$
- For any $\omega \in \widetilde{\Omega}$ such that $P(\widetilde{\Omega}) = 1$, we have that $B(\omega, -) : T \to S$ is a continuous function.

4 Martingale Processes

4.1 Basic Definitions

Definition 4.1.1 (Martingale Process) Let (Ω, E, P) be a probability space. Let (S, Σ) be a measurable space. Let $M : \Omega \times [0, T] \to S$ be a continuous stochastic process adapted to a filtration $\mathcal{F} = \{\mathcal{F}_t \mid t \in [0, T]\}$. We say that M is Martingale with respect to $\mathcal{F}$ if the following are true.

- $E[|M_k|] < \infty$ for all k.
- $E[M_{t_2} \mid \mathcal{F}_{t_1}] = M_{t_1}$ for all $0 \leq t_1 < t_2 \leq T$.

Definition 4.1.2 (Continuous Martingale Process) Let (Ω, E, P) be a probability space. Let (S, Σ) be a measurable space. Let $M : \Omega \times [0, T] \to S$ be a continuous stochastic process. We say that M is continuous if for all $\omega \in \widetilde{\Omega}$ such that $P(\widetilde{\Omega}) = 1$, the function

$$M(\omega, -) : [0, T] \to S$$

is continuous.

Definition 4.1.3 (Standard Brownian Filtration)

4.2 Martingale Transformations

5 Stochastic Calculus

5.1 Ito's Integral

Given a Martingale process $M : \Omega \times [0, T] \to \mathbb{R}$ on some filtered probability space, we may ask to integrate M over the product measure of the measurable space $\Omega \times [0, T]$ because M is in particular a measurable function. Ito's integral gives a construction of such an integral under certain assumption on M.

Definition 5.1.1 (Simple Functions) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Define $\mathcal{H}_0^2 \subseteq L^2(\Omega \times [0, T])$ to be the vector subspace whose measurable functions are of the form

$$f(\omega, t) = \sum_{i=0}^{n-1} a_i(\omega) 1_{(t_i, t_{i+1}]}$$

Definition 5.1.2 (Ito Integral of Simple Functions) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $f \in \mathcal{H}_0^2$. Define the Ito integral of $f(\omega, t) = \sum_{i=0}^{n-1} a_i(\omega) 1_{(t_i, t_{i+1}]}$ by the formula

$$I(f)(\omega) = \sum_{t=0}^{n-1} a_i(\omega)(B_{t_{i+1}} - B_{t_i})$$

In particular, I is a function $I : \mathcal{H}_0^2 \to L^2(\Omega)$.

Lemma 5.1.3 (Ito's Isometry I) Let $(\Omega, \mathcal{F}, P)$ be a probability space. The map $I : \mathcal{H}_0^2 \to L^2(\Omega)$ is a linear isometry.

Definition 5.1.4 (Ito Integrable Functions) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Define the space of Ito integrable functions $\mathcal{H}^2 \subseteq L^2(\Omega \times [0, T])$ to be the vector subspace whose measurable functions $f \in L^2(\Omega \times [0, T])$ satisfies the fact that

$$E\left[\int_{[0,T]} f^2(\omega, t)\, dt\right] = \int_\Omega \int_{[0,T]} f^2(\omega, t)\, dt\, dP < \infty$$

Lemma 5.1.5 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Then $\mathcal{H}_0^2$ is dense in $\mathcal{H}^2$.

Definition 5.1.6 (Ito Integral of Integrable Functions) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $f \in \mathcal{H}^2$. Let $(f_n)_{n \in \mathbb{N} \setminus \{0\}}$ be a sequence such that $f_n \to f$ in $L^2(\Omega \times [0, T])$. Then define the Ito integral of f by

$$I(f) = \lim_{n \to \infty} I(f_n)$$

In particular, I is a function $I : \mathcal{H}^2 \to L^2(\Omega)$.

Lemma 5.1.7 (Ito's Isometry II) Let $(\Omega, \mathcal{F}, P)$ be a probability space. The map $I : \mathcal{H}^2 \to L^2(\Omega)$ is a linear isometry.

5.2 The Integral as a Martingale

5.3 Ito's Lemma

5.4 Localization

6 Stochastic Differential Equations

6.1 Basic Definitions

Definition 6.1.1 (Stochastic Differential Equations)

Methods of solving differential equations can also be applied to stochastic differential equations.

6.2 Geometric Brownian Motions

Definition 6.2.1 (Geometric Brownian Motion) Let (Ω, E, P) be a probability space. Let $S : \Omega \times [0, T] \to \mathbb{R}$ be a continuous Martingale process with respect to the standard Brownian filtration $\mathcal{B} = \{B_t \mid t \in [0, T]\}$. We say that S follows a geometric Brownian motion if it satisfies the following stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

where μ is called the percentage drift and σ is called the percentage volitality.

Recall that dS_t really means the Radon–Nikodym derivative $\frac{dS_t}{d\mu}$ where μ here (not the same one as above) refers to the standard measure on $\mathbb{R}$, while dt means the genuine infinitesimally small increment in $\mathbb{R}$.

Lemma 6.2.2 Let (Ω, E, P) be a probability space. Let $S : \Omega \times [0, T] \to \mathbb{R}$ be a continuous Martingale process with respect to the standard Brownian filtration $\mathcal{B} = \{B_t \mid t \in [0, T]\}$. Suppose that S follows a geometric Brownian motion. Then the solution of the stochastic differential equation is given by

$$S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma B_t}$$

Proposition 6.2.3 Let (Ω, E, P) be a probability space. Let $S : \Omega \times [0, T] \to \mathbb{R}$ be a continuous Martingale process with respect to the standard Brownian filtration $\mathcal{B} = \{B_t \mid t \in [0, T]\}$. Suppose that S follows a geometric Brownian motion. Then the following are true.

- For each t, S_t has a log normal distrbution.
- For each t, we have
$$E[S_t] = S_0 e^{\mu t}$$
- For each t, we have
$$\text{Var}(S_t) = S_0^2 e^{2\mu t}(e^{\sigma^2 t} - 1)$$

We also provide a discrete approach to solving the differential equation for easy simulation using the Monte Carlo method.

Proposition 6.2.4 The discretization of the defining stochastic differential equation of a geometric Brownian motion is given by

$$S_{t+dt} - S_t = \mu S_t dt + \sigma S_t N \sqrt{dt}$$

where dt denotes a small change in time, and $N \sim N(0, 1)$ refers a random variable with a normal distribution.

6.3 Systems of SDEs